

Proof of convergence

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When given a square chunk of dimension $n \times n$, the recursive process will do one of two things: if n is even, it will split the chunk into four chunks whose dimension is half the original. If n is odd, it will split the chunk into an "L" shaped block and another square chunk of dimension $(n-1) \times (n-1)$.

Here we will prove that this is enough to generate a mapping for an image of any Integer dimension n .

Assume a function $f(n)$ that tracks the dimension of the next square chunk based on the current chunk. If we can show that this must be equal to one at some point, then we are done because this shows that the algorithm will individually encompass all points at time iteration. The exact form of f is given below.

$$f(x) = \begin{cases} x-1 & \text{if } x \equiv 1 \pmod{2} \\ \frac{x}{2} & \text{if } x \equiv 0 \pmod{2} \end{cases}$$

This is simply derived from the above description of the recursive algorithm. We will proceed with a proof by contradiction. Repeated applications of f and n number of times will be denoted f^n .

Note, f^n is monotonically decreasing on n , that is to say $f^i(x) \leq f^j(x)$ if $i \geq j$.

Suppose there is some special value $\delta \in \mathbf{N}$ such that no matter how many times f is applied, its value always remains above 0. There will exist some smallest value $\gamma > 0$ that we will assume it reaches at $n = n_\gamma$.

$$f^{n_\gamma}(\delta) = \gamma$$

$$f^k(\delta) \geq \gamma \forall k \neq n_\gamma$$

Now consider the value of $f(\gamma)$. This is equivalent to $f^{n_\gamma+1}(\delta)$, which means it cannot be greater than γ because f is monotonically decreasing on n . However, it also cannot be less than γ because γ is the smallest value of f on δ by definition. therefore,

$$f(\gamma) = \gamma$$

Looking at the two branches of f this is only possible if γ is even in which case $\gamma = \frac{\gamma}{2}$ which necessarily means that $\gamma = 0$ which is a direct contradiction with the first statement.

This shows that there must exist some finite integer value of n where

$$\exists n; f^n(x) = 0 \forall x \in \mathbf{N}$$

Since $f^0(x) = x > 0$ this means there is some first value of n where $f^n(x) = 0$ for all x . Lets call this value n_x^* . Consider

$$f^{n_x^*-1}(x) = f^{-1}(f^{n_x^*}(x)) = f^{-1}(0)$$

Based on the definition of f , the only possible inverse values for 0, are 0 itself and 1. However, $f^{n_x^*-1}(x) \neq 0$ because n_x^* is by definition the first n value where $f^n(x) = 0$. This means that $f^{n_x^*-1}(x) = 1$ for all x . Thus proving that successive splits using our algorithms recursive process will eventually cover every pixel.